

Liang-Hsuan Tseng (曾亮軒)

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ABOUT ME

I am a second-year Ph.D. student at the Speech Processing and Machine Learning (SPML) Laboratory of National Taiwan University, instructed by Prof. Hung-yi Lee. My research primarily focuses on speech processing and applied machine learning, with particular interests in automatic speech recognition and spoken language modeling. I have developed a subtitle generator system that automatically transcribes lecture recordings for the entire university, significantly improving accessibility in education. Recently, I have been working on spoken language modeling, with the aim of developing intelligent and high-fidelity speech generative models.

EDUCATION

- National Taiwan University** 2022-present
Ph.D. in Graduate Institute of Electrical Engineering GPA: 4.14 / 4.30
- National Taiwan University** 2018-2022
Bachelor of Electrical Engineering GPA: 3.82 / 4.30

EXPERIENCE

- MediaTek Research** 2024-2025
Deep Learning Intern Taipei, Taiwan
 - Propose TASTE: Text-Aligned Speech Tokenization and Embedding for Spoken Language Modeling [ICLR'26]
 - The first hybrid text-speech tokenization tailored for spoken language modeling.
- Speech Processing and Machine Learning Laboratory, National Taiwan University** 2022-present
Ph.D. Student / Advisor: Prof. Hung-yi Lee Taipei, Taiwan
 - Derive SOTA unsupervised ASR framework based on reinforcement learning and iterative training. [NeurIPS'24]
 - Improve code-switching automatic speech recognition (ASR) with self-supervised speech models. [AAAI SAS'22]
- Center of Teaching and Learning Development, National Taiwan University** 2021-present
Research and Development Intern Taipei, Taiwan
 - Develop and deploy the course subtitle generator for National Taiwan University.
 - Improve the ASR model of the subtitle generator. The new model is 2 times smaller, 5 times faster, and 20% better than the Whisper Large-v2 in code-switching ASR. [SLT'24]
 - Our system has transcribed over 80,000 lecture videos for the university faculty.

PUBLICATIONS

- TASTE: Text-Aligned Speech Tokenization and Embedding for Spoken Language Modeling** [ICLR'26]
Liang-Hsuan Tseng, Yi-Chang Chen, Kuan-Yi Lee, Da-Shan Shiu, Hung-yi Lee
- REBORN: Reinforcement-Learned Boundary Segmentation with Iterative Training for Unsupervised ASR** [NeurIPS'24]
Liang-Hsuan Tseng, En-Pei Hu, Cheng-Han Chiang, Yuan Tseng, Hung-yi Lee, Lin-shan Lee, Shao-Hua Sun
- Leave No Knowledge Behind during Knowledge Distillation: Towards Practical and Effective Knowledge Distillation for Code-Switching ASR Using Realistic Data** [SLT'24]
Liang-Hsuan Tseng, Zih-Ching Chen, Wei-Shun Chang, Cheng-Kuang Lee, Tsung-Ren Huang, Hung-yi Lee
- Improving cascaded unsupervised speech translation with denoising back-translation** [Preprint]
Yu-Kuan Fu, Liang-Hsuan Tseng (co-first), Jiatong Shi, Chen-An Li, Tsu-Yuan Hsu, Shinji Watanabe, Hung-yi Lee
- Mandarin-English Code-Switching Speech Recognition with Self-Supervised Speech Representation Models** [AAAI SAS'22]
Liang-Hsuan Tseng, Yu-Kuan Fu, Heng-Jui Chang, Hung-yi Lee
- On the Fallacy of Global Token Perplexity in Spoken Language Model Evaluation** [ACL Findings'26]
Jeff Chan-Jan Sju, Liang-Hsuan Tseng, Yi-Cheng Lin, Yen-Chun Kuo, Ju-Chieh Chou, Kai-Wei Chang, Hung-yi Lee, Carlos Busso

- **A Self-Refining Framework for Enhancing ASR Using TTS-Synthesized Data** [ASRU'25]
Cheng-Kang Chou, Chan-Jan Hsu, Ho-Lam Chung, Liang-Hsuan Tseng, Hsi-Chun Cheng, Yu-Kuan Fu, Kuan Po Huang, Hung-Yi Lee
- **Introducing Semantics into Speech Encoders** [ACL'23]
Derek Xu, Shuyan Dong, Changhan Wang, Suyoun Kim, Zhaojiang Lin, Akshat Shrivastava, Shang-Wen Li, Liang-Hsuan Tseng, Alexei Baevski, Guan-Ting Lin, Hung-yi Lee, Yizhou Sun, Wei Wang
- **Improving Generalizability of Distilled Self-Supervised Speech Processing Models under Distorted Settings** [SLT'22]
Kuan-Po Huang, Yu-Kuan Fu, Tsu-Yuan Hsu, Fabian Ritter Gutierrez, Fan-Lin Wang, Liang-Hsuan Tseng, Yu Zhang, Hung-yi Lee

ACADEMIC SERVICE

- **Reviewer:** ACL ARR (multiple times), NeurIPS 2025, TASLP 2025, ICLR 2026, ICML 2026
- **Emergency Reviewer:** ASRU 2025, ACML 2025, ICLR 2026, Interspeech 2026

INVITED TALKS

- **REBORN Unsupervised ASR System** Winter, 2024
A 1 hour talk about our SOTA UASR system (NeurIPS 2024) Massachusetts Institute of Technology
– Invited by Spoken Language Systems group at MIT CSAIL

HONORS AND AWARDS

- **NSTC Graduate Research Fellowship** 2025-2028
National Ph.D. fellowship NSTC, Taiwan
- **Outstanding Doctoral Talent Cultivation Program** 2026
Institutional Ph.D. fellowship GICE, National Taiwan University
- **NVIDIA Academic Grant Program** 2026
Computing Grant for Spoken Language Modeling Development NVIDIA
- **Science and Technology Research Innovation Award** 2024
Research award of our SOTA UASR system published in NeurIPS 2024 National Taiwan University
- **Conference Travel Grant from the Foundation for the Advancement of Outstanding Scholarship** 2024
Conference travel grant to NeurIPS 2024 FAOS, Taiwan

TEACHING EXPERIENCE

- **Machine Learning 2023** Spring 2023
Head Teaching Assistant / Instructor: Prof. Hung-yi Lee National Taiwan University
– The head TA of a course with over 500 students and 20 teaching assistants
– Manage and examine the design and grading of all the 15 assignments
- **Deep Learning for Human Language Processing 2022** Fall 2022
Teaching Assistant / Instructor: Prof. Hung-yi Lee National Taiwan University
– Design and provide instructions for the assignment of unsupervised ASR
- **Machine Learning 2022** Spring 2022
Teaching Assistant / Instructor: Prof. Hung-yi Lee National Taiwan University
– Design and provide instructions for the assignment of model compression

SKILLS

- **Programming:** Python, JavaScript, C/C++, C#, L^AT_EX
- **Tools & Frameworks:** Pytorch, Docker, Node.js, React, PostgreSQL, RabbitMQ